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Department of CSE

B.Tech. CSE Trimester -5

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Computer Network and Networks Programming

25CS2208E

SKILL RECORD

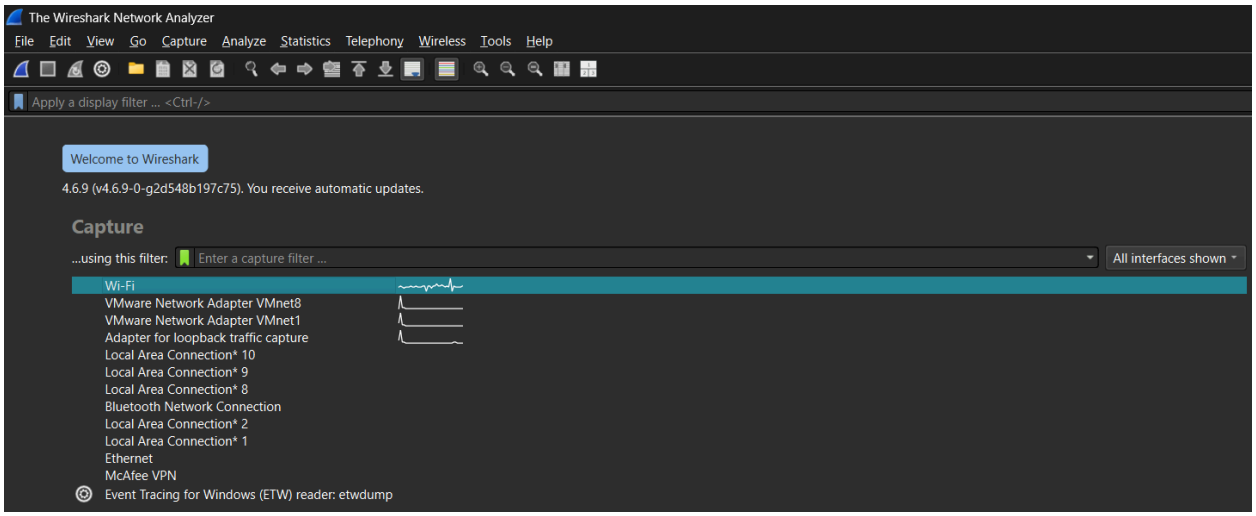
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Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

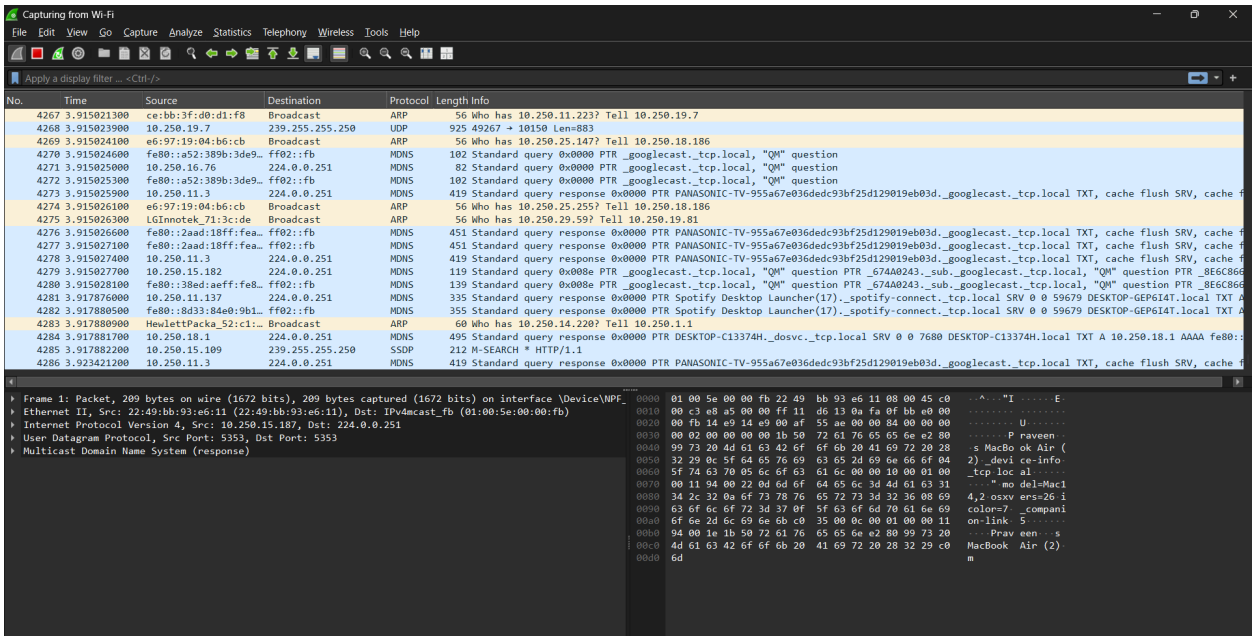
1. Open the browser.
2. Install Wireshark [Wireshark • Go Deep | Download]
3. Install it using the default settings.
4. Open Wireshark.



Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as: Wi-Fi, Ethernet, Bluetooth

1. If you are connected through Wi-Fi, select Wi-Fi.
2. Double-click the Wi-Fi interface.
3. Packet capture will start.



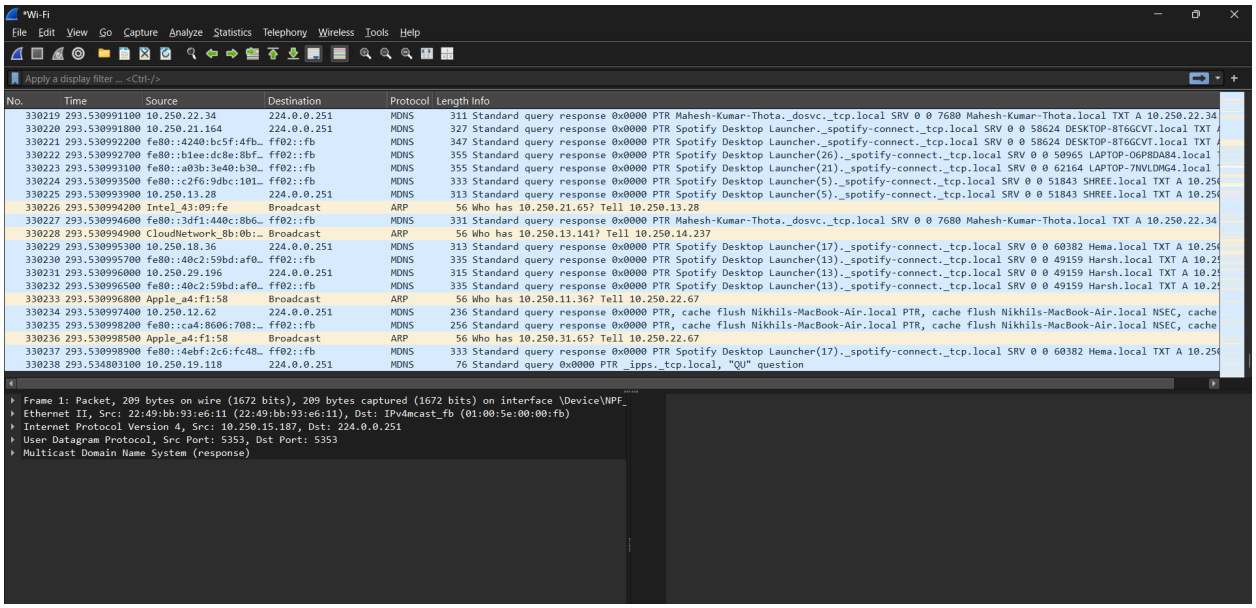
Step 3: Generate network traffic

While Wireshark is capturing:

1. Open a new browser tab.

2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

You should now see captured packets.



Step 4: Apply display filters

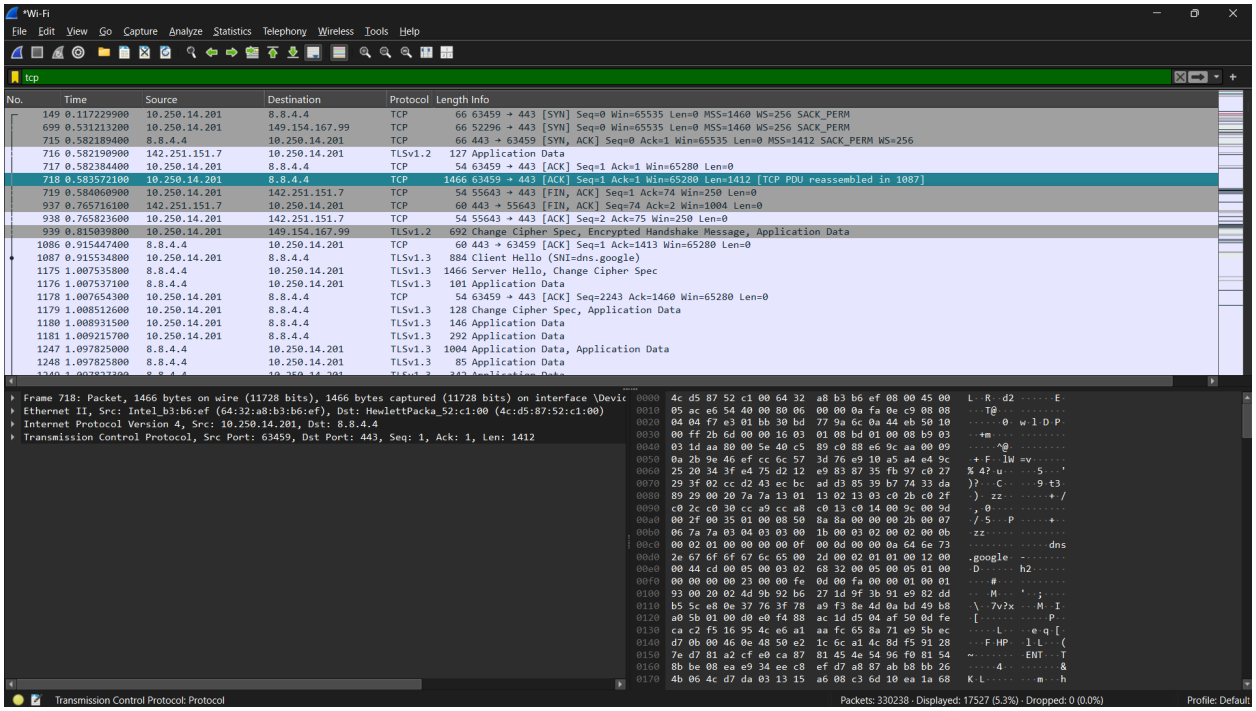
In the Wireshark filter bar, enter each filter separately and press Enter.

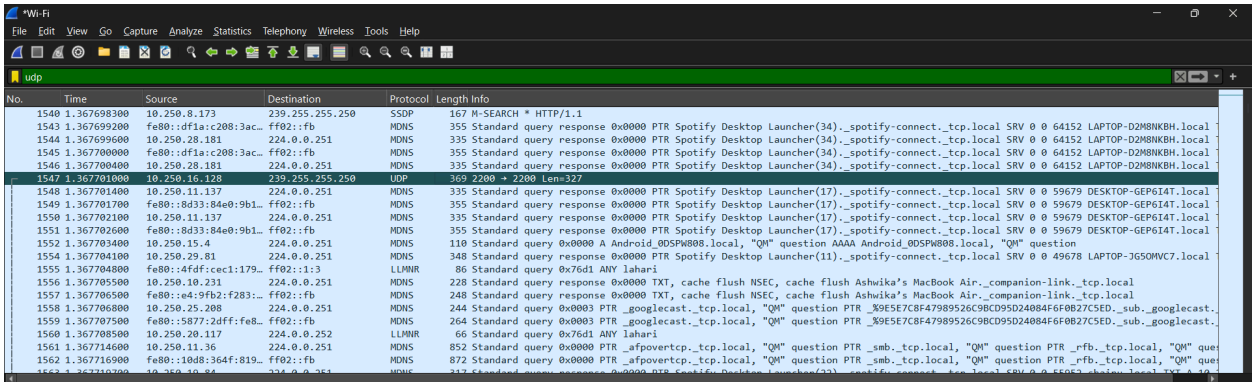
Filter	Purpose
tcp	Displays TCP packets
udp	Displays UDP packets
icmp	Displays ICMP packets
tcp.len > 0	Displays TCP packets carrying data

First, enter: TCP

Observe the packets displayed. Then enter: UDP

Observe the UDP packets.





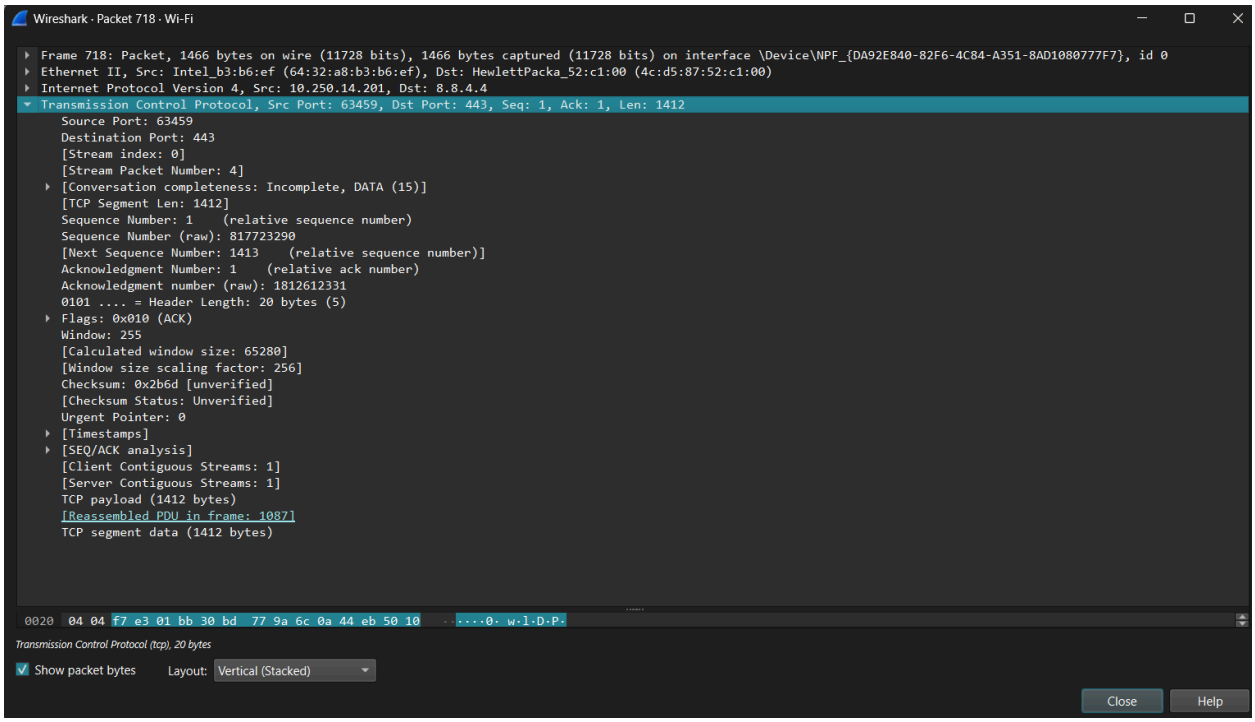
Step 5: Inspect Packet Headers

Write the following:

Select any TCP packet from the packet list.

In the packet details pane, expand these sections:

1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol



Parameter	**TCP Packet**
Frame Length	1466 bytes
Ethernet Header	14 bytes
Source MAC Address	64:32:a:8:b3:b6:ef
Destination MAC Address	4c:d5:87:52:c1:1:00
IPv4 Header Length	20 bytes
IPv4 Total Length	1452 bytes
TCP Header Length	20 bytes
TCP Segment Length (Payload)	1412 bytes
TCP Segment Total Length	1432 bytes

Parameter	**UDP Packet**
Frame Length	369 bytes
Ethernet Header	14 bytes
Source MAC Address	a2:c6:15:10:57:3a
Destination MAC Address	01:00:5e:7f:ff:fa
IPv4 Header Length	20 bytes
IPv4 Total Length	355 bytes
UDP Header Length	8 bytes
UDP Length	335 bytes
UDP Payload Length	327 bytes

Step 6: Calculate Header Overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Application data = 1500 bytes
- FCS = 4 bytes, if included in the frame representation

Total size including FCS:

$$14 + 20 + 20 + 1500 + 4 = 1558bytes$$

Total overhead:

$$14 + 20 + 20 + 4 = 58bytes$$

Payload efficiency:

1500/1558 *100 = 96.28%

- UDP Length = 335 bytes
- UDP Payload = 327 bytes

Therefore:

$$8 + 327 = 335bytes$$

The UDP length includes both the UDP header and its payload.

Step 7:

Fill in your observation table

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	1466 bytes	369 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	1452 bytes	355 bytes
Transport Segment Length	1432 bytes	335 bytes
Payload Length	1412 bytes	327 bytes